CRICKET SCORE PREDICTION USING MACHINE LEARNING

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PROJECT OVERVIEW & OBJECTIVES

- **Objective:** Build a Real-Time Machine Learning System to predict ODI Cricket Scores.
- **The Approach:** Unlike static run-rate calculators, this system uses Progressive Predictive Modeling (Ball-by-ball analysis) so as the match progresses it continuously updates with match state and improves accuracy.
- **Key Capabilities:** Real-time prediction, Context awareness (Venue/Pressure), and "What-If" scenario testing.

THE PROBLEM & THE PIVOT

- **The Core Problem:** Traditional predictors rely on Current Run Rate, which fails because it treats 120/0 (Dominant) and 120/5 (Collapse) as the exact same scenario. To distinguish these scenarios, a model must understand Player Quality (Who is left?) and Venue History (Is this a batting track?) so it understands context for which it requires real calculated data which our V1 was lacking in.

Data Component	v1 (Context-Blind)	v2 (Context-Aware)
Player Logic	Hardcoded Default (35) We filled missing data with a generic "35 runs." The model couldn't see a collapse because it thought every tail-ender was a top-order batsman.	Role-Based Calculation We calculated real stats and assigned roles: 18 (Bowler) vs 30 (Batsman). Now the model sees the weak tail.
Venue Logic	Hardcoded Default (250) Assumed every pitch was "average," so the model couldn't learn how pitch conditions dictate the score ceiling.	Historical Intelligence Calculated real historical averages for 303 specific venues, teaching the model to adjust for pitch types.
Data Component	v1 (Context-Blind)	v2 (Context-Aware)

DATA INTEGRITY & VALIDATION STRATEGY

Metric	v1 (Flawed Approach)	v2 (Rigorous Approach)	Why it matters?
Data Quality	Mixed Quality(Included noisy Domestic/Club matches)	International Only(Filtered for ICC Full Members)	Removes Noise: Domestic stats vary wildly; International data is consistent .
Validation Method	Random Split (80/20) (Shuffled matches randomly)	Temporal Split(Strict Time-Based Separation)	Prevents Cheating: Random splits leak future trends. Temporal simulates real life.
Dataset Split	Black Box(Unknown distribution)	Training: 2002–2022 (11,064 Matches) Testing: 2023–2025 (257 Matches)	Zero Leakage: The model predicts 2023 matches having never seen them during training.

COMPARISON OF BOTH VERSIONS ON SAME DATASET

Metric	v1 (Previous)	v2 (Final)	Improvement
Overall R^2 Score	0.527	0.613	+16% Accuracy
Error (MAE)	37.8 runs	29.2 runs	-23% Error (8.6 runs less)
Death Overs R^2	0.873	0.941	+7.8% (Near-Perfect)
Death Overs MAE	18.0 runs	17.2 runs	-4.4% Error
Weak Teams R^2	0.412	0.514	+25% Better

MODEL SELECTION (RANDOM FOREST VS XGBOOST)

Stage	XGB R^2	XGB MAE	RF R^2	RF MAE	Insight
Pre-Match	0.305	41.9 Runs	0.158	47.1 Runs	XGB Wins: Better at guessing with limited info.
Early (Over 10)	0.570	31.4 Runs	0.533	34.2 Runs	XGB Wins: Captures linear Run-Rate logic best.
Mid (Over 20)	0.714	25.5 Runs	0.718	26.6 Runs	Tipping Point: Models tied; RF starts catching up.
Late (Over 30)	0.821	19.8 Runs	0.837	20.2 Runs	RF Leads: Wickets/Complexity start dictating score.
Death (Over 40)	0.923	12.7 Runs	0.941	11.7 Runs	RF Dominates: Perfect at detecting "cliff edge" collapses.

Key Decision: We prioritized Random Forest because achieving 94% accuracy (Error ~11 runs) in the Death Overs is far more valuable to users than early-game estimations.

REAL MATCH VALIDATION

Case Study: Match: India vs Australia **Venue: M. Chinnaswamy Stadium (avg 298)**

Stage	Score	v1 Prediction	v2 Prediction	Actual Final
Pre-match	0/0	295	310	352
Early (10 ov)	70/0	318	335	352
Mid (20 ov)	145/1	342	348	352
Late (30 ov)	210/2	335	351	352
Death (40 ov)	290/3	351	354	352
Result	-	Error: 17	Error: 1	352

REAL MATCH VALIDATION IN COLLAPSE

Match Date: September 2023   Venue: Gaddafi Stadium (avg 252)

Stage	Score	Wkts	Model Prediction	Actual Final	Error	Model Reaction
Mid (20 ov)	110/2	2	278	228	+50	On track for high score
Collapse (30 ov)	110/6	6	240	228	+12	-38 Run Correction
Death (40 ov)	180/7	7	226	228	-2	Stabilized

Key Insight: Model identified the collapse instantly. When wickets fell from 2→6, prediction dropped from 278 → 240 (38-run adjustment). Final error: 2 runs.

Comparison with v1: v1 would have kept predicting ~260-265 (only -15 run adjustment), leading to ~30+ run error. v2's aggressive -38 run adjustment shows it understands "4 wickets quickly = tail exposed = severe impact.

"WHAT-IF" ANALYSIS (CONTEXT AWARENESS)

Does the model understand Pressure & Player Quality?

Scenario	Situation	Adding Virat Kohli	Impact	Sensitivity
Crisis	60/4 (15 ov)	Prediction: 218	+33 Runs	HIGH
Comfort	180/2 (30 ov)	Prediction: 298	+15 Runs	LOW
Tail-Ender	220/7 (42 ov)	Adding All-Rounder: 268	+26 Runs	MEDIUM

The model recognizes that elite players are more valuable under pressure—Kohli's impact drops from +33 runs in a crisis to +15 runs when the team is comfortable, proving contextual awareness.

WHAT IF: VENUE IMPACT ANALYSIS

Match State	Venue	Prediction	Difference
India 150/3	M. Chinnaswamy (Batting Paradise)	325	Baseline
India 150/3	Eden Gardens (Balanced)	305	-20 runs
India 150/3	Mirpur (Bowling-Friendly)	277	-48 runs

Real Match Validation:

- India 150/3 at Chinnaswamy → Actual: 321, Model: 325 (error: 4 runs)
- India 148/3 at Mirpur → Actual: 273, Model: 277 (error: 4 runs)

Key Insight:

Same match state (150/3) produces a 48-run difference based purely on venue characteristics—proving the model understands pitch conditions.

WHAT-IF ANALYSIS: MATCH STATE & MOMENTUM

Scenario	Match State	Prediction	Impact	Insight
High Momentum	200/4 (35 ov), 90 runs last 10	295	Baseline	Set batsmen, acceleration coming
Stagnation	200/4 (35 ov), 40 runs last 10	262	-33 runs	Model recognizes scoring struggle
Cluster Wickets	180 runs (30 ov), lost 3→6 wickets	258	-52 runs	Tail exposed, panic mode detected

Key Findings: Model Logic

- Momentum: A 15-run shift in the last 10 overs impacts the final score by ~15 runs.
- Panic Detection: Losing 3 wickets in a cluster drops the prediction by 52 runs instantly .
- Non-Linearity: The exact same score (180 runs) varies by 68 runs depending on wickets lost .

Real Match Validation

- Momentum Test: India (High Momentum) -> Predicted 295, Actual 299 (Error: 4 runs) .
- Collapse Test: Sri Lanka (Panic at 181/6) → Predicted 255, Actual 248 (Error: 7 runs) .

LIMITATIONS & ERROR ANALYSIS

Error Source	Frequency	Why it fails?
Unexpected Collapses	40%	Human Factor: Cannot foresee a "Brain Fade" (e.g., losing 4 wickets for 10 runs). R2 drops to 0.42 in these chaos scenarios.
Individual Brilliance	30%	Statistical Outliers: Cannot predict a tail-ender scoring a random 50 runs or uncharacteristic partnerships.
Extreme Scores	20%	Data Scarcity: Scores >400 or <150 are rare events. The model has less training data for these extremes.
Pre-Match Uncertainty	10%	Lack of Info: Before Ball 1, R2 is only 0.26. The model cannot see pitch behavior or weather yet.

Performance Gaps:

- Domestic Blind Spot: Model performs 21% worse (R2 0.45) on domestic matches due to inconsistent player quality and data noise.

Key Takeaway:

- Only 3.9% of predictions result in catastrophic error (>100 runs), almost entirely due to the unpredictable scenarios listed above

CONCLUSION

What We Accomplished:

- **Engineered Realism:** Transformed a static calculator into a Context-Aware System by parsing 2,800 matches to calculate real venue and player logic.
- **Precision Where It Counts:** Achieved 0.94 R^2 (94% Accuracy) in the critical Death Overs using our Random Forest Champion model.
- **Validated Intelligence:** Proved the model understands "Game Flow"—valuing elite players more during collapses and adjusting for specific venue history.
- **Full-Stack Product:** Delivered a fully functional React.js Dashboard where users can interactively test "What-If" scenarios and simulate matches in real-time.

THANK YOU

KING KRLEGA!